

Algorithm for Preparing Thermal Gibbs States

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August 4, 2010

This work is based on

- *Sampling from the Thermal Quantum Gibbs State and Evaluating Partition Functions with a Quantum Computer*
with David Poulin, Phys. Rev. Lett. 103, 220502 (2009)
- *Algorithm for Preparing Thermal States – Detailed Analysis*
with C.-F. Chiang, Proc. NATO ARW “Quantum Cryptography and Computing: Theory and Implementation, Gdansk, Poland, 9-12 September 2009

This work has been supported by the National Science Foundation grants CCF-0726771 and CCF-0746600

Let H be a Hamiltonian with spectral decomposition

$$H = \sum_{a=1}^D E_a |\psi_a\rangle \langle \psi_a|$$

The partition function Z_β at inverse temperature β is

$$Z_\beta := \sum_{a=1}^D e^{-\beta E_a}$$

The thermal state ρ_β at inverse temperature β is

$$\rho_\beta := \sum_{a=1}^D \frac{e^{-\beta E_a}}{Z_\beta} |\psi_a\rangle \langle \psi_a|$$

Thermalizing quantum systems

Given H , β , and ϵ , prepare a quantum state $\tilde{\rho}_\beta$ with

$$\|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \epsilon$$

Metropolis Sampling

Quantized Metropolis Sampling

- Somma, Boixo, Barnum, and Knill, PRL 101,130504
- W. and Abeyesinghe, PRA 78, 032311

Method based on Amplitude Amplification (this presentation)

Quantum Metropolis Sampling (David Poulin's presentation)

Let Ω be a set whose elements x correspond to the states of some classical system

Let $E : \Omega \rightarrow [0, 1]$ denote the energy function, assigning to each state x its energy $E(x)$

Let

$$H = \sum_{x \in \Omega} E(x) |x\rangle \langle x|$$

Thermalizing such classical systems is easier because H is diagonal in the computational basis $\{|x\rangle\}_{x \in \Omega}$

Metropolis sampling

Given such classical Hamiltonian H , we can always construct a Markov chain P whose limiting distribution is equal to ρ_β

The spectral gap δ determines how fast we can approach the Boltzmann distribution

The run time scales like $1/\delta$

In general, it is not possible to determine (or bound) the spectral gap $\Rightarrow$ heuristics

There are important cases where it can be shown that the gap is large

Quantized Metropolis algorithm

Given such classical Hamiltonian H , there is a “quantized” Metropolis algorithm preparing a state $|\tilde{\varphi}_\beta\rangle$ with

$$\| |\varphi_\beta\rangle - |\tilde{\varphi}_\beta\rangle \| \leq \epsilon$$

where $|\varphi_\beta\rangle$ is a “quantized” version of the Boltzmann distribution

$$|\varphi_\beta\rangle := \sum_{x \in \Omega} \sqrt{\frac{e^{-\beta E_x}}{Z_\beta}} |x\rangle$$

The run times scales like $\sqrt{1/\delta}$

Structure of our method

Prepare the state

$$\frac{1}{\sqrt{D}} \sum_{x \in \Omega} |x\rangle \otimes |E_x\rangle \otimes |0\rangle$$

Apply a controlled rotation to obtain the state

$$|\psi\rangle = \frac{1}{\sqrt{D}} \sum_{x \in \Omega} |x\rangle \otimes |E_x\rangle \otimes \left(\sqrt{e^{-\beta E_x}} |0\rangle + \sqrt{1 - e^{-\beta E_x}} |1\rangle \right)$$

Structure of our method

Measure in the basis $|0\rangle, |1\rangle$

If we obtain $|0\rangle \Rightarrow$ the post-measurement state is

$$\frac{|\Pi_0|\Psi\rangle}{\|\Pi_0|\Psi\rangle\|} = \sum_{x \in \Omega} \sqrt{\frac{e^{-\beta E_x}}{Z_\beta}} |x\rangle \otimes |E_x\rangle \otimes |0\rangle$$

where $\Pi_0 = I \otimes |0\rangle\langle 0|$

Using amplitude amplification, we can achieve an expected run time

$$\frac{1}{\|\Pi_0|\Psi\rangle\|} = \sqrt{\frac{D}{Z_\beta}}$$

Generalization to quantum Hamiltonians

Let H be an arbitrary quantum Hamiltonian with spectral decomposition

$$H = \sum_{a=1}^D E_a |\psi_a\rangle \langle \psi_a| \quad E_a \in [0, 1/2]$$

There are two difficulties:

- we cannot change between the computational basis $|a\rangle$ and the eigenvector basis $|\psi_a\rangle$ of H
- we do not know the energies E_a

Both difficulties can be overcome with quantum phase estimation
 $\Rightarrow$ we can extend the previous method to quantum Hamiltonians

Given H , β , and ϵ , our quantum algorithm prepares $\tilde{\rho}_\beta$

$$\|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \epsilon$$

The errors come from two sources:

$$\|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \|\rho - \rho_{\text{sim}}\|_{\text{tr}} + \|\rho_{\text{sim}} - \rho_{\text{fail}}\|_{\text{tr}}$$

- ρ_{sim} due to imperfect simulation of $\exp(2\pi i H)$
- $\tilde{\rho}_\beta = \rho_{\text{fail}}$ due to nonzero failure probability and limited precision of quantum phase estimation

Imperfect simulation of Hamiltonians

Assume that H is sparse and let $U = \exp(2\pi iH)$

Using techniques for simulating Hamiltonian time evolutions, we can implement U_{sim} with

$$\|U - U_{\text{sim}}\| \leq \epsilon_{\text{sim}}$$

The cost scales like

$$\text{poly}(\log D, 1/\epsilon_{\text{sim}}, s)$$

where s indicates the sparseness of H

Matrix logarithm is Lipschitz continuous:

$$\|U - U_{\text{sim}}\| \leq \epsilon_{\text{sim}}$$

$\Rightarrow$ there is a Hamiltonian H_{sim} with

$$\|H - H_{\text{sim}}\| \leq O(\epsilon_{\text{sim}})$$

Thermal states are Hölder continuous:

$$\|H - H_{\text{sim}}\| \leq O(\epsilon_{\text{sim}})$$

$\Rightarrow$ the corresponding thermal states satisfy

$$\|\rho - \rho_{\text{sim}}\|_{\text{tr}} \leq O(\sqrt{\beta \epsilon_{\text{sim}}})$$

Structure of our method

Start with the maximally entangled state

$$|\Phi\rangle := \frac{1}{\sqrt{D}} \sum_{a=1}^D |a\rangle \otimes |a\rangle$$

Run phase estimation of U_{sim} on first part of $|\Phi\rangle$ and record eigenphase in the energy register

Apply the controlled rotation $R = \sum_E |E\rangle\langle E| \otimes R_E$

$$R_E = \begin{pmatrix} \sqrt{e^{-\beta E}} & -\sqrt{1 - e^{-\beta E}} \\ \sqrt{1 - e^{-\beta E}} & \sqrt{e^{-\beta E}} \end{pmatrix}$$

Use amplitude amplification to increase overlap with $|0\rangle$ component

Invariance of maximally entangled state

Let

$$|\Phi\rangle := \frac{1}{\sqrt{D}} \sum_{a=1}^D |a\rangle \otimes |a\rangle$$

$$V := \sum_{a=1}^D |\psi_a\rangle \langle a|$$

Due to invariance of $|\Phi\rangle$ under $V \otimes \bar{V}$, we have

$$|\Phi\rangle = \frac{1}{\sqrt{D}} \sum_{a=1}^D |\psi_a\rangle \otimes |\bar{\psi}_a\rangle$$

Correspondance between energy levels and eigenphases

Recall that $U = \exp(2\pi iH)$

We have

$$(U \otimes I_D)|\psi_a\rangle \otimes |\bar{\psi}_a\rangle = e^{2\pi i E_a} |\psi_a\rangle \otimes |\bar{\psi}_a\rangle$$

$$(U \otimes I_D)|\Phi\rangle = \frac{1}{\sqrt{D}} \sum_{a=1}^D e^{2\pi i E_a} |\psi_a\rangle \otimes |\bar{\psi}_a\rangle$$

The energy levels E_a of H correspond bijectively to the eigenphases E_a of U

$\Rightarrow$ Apply quantum phase estimation to approximate the energy levels

Intuition – perfect quantum phase estimation

If perfect quantum phase estimation were possible (infinite precision and zero probability of failure), we could prepare the state

$$|\Psi\rangle = \frac{1}{\sqrt{D}} \sum_{a=1}^D |\psi_a\rangle \otimes |\bar{\psi}_a\rangle \otimes |E_a\rangle \otimes \left(\sqrt{e^{-\beta E_a}} |0\rangle + \sqrt{1 - e^{-\beta E_a}} |1\rangle \right)$$

⇒ Using amplitude amplification just as in the case of classical Hamiltonians, we could obtain a purification of the thermal state

Imperfect quantum phase estimation

We use quantum phase estimation that invokes the controlled U_{sim} operation

$$O((1/\epsilon_{\text{prec}}) \log(1/\epsilon_{\text{fail}})) \text{ times}$$

$\Rightarrow$ We transform $|\Phi\rangle$ to the state

$$\frac{1}{\sqrt{D}} \sum_{a=1}^D |\psi_a\rangle \otimes |\bar{\psi}_a\rangle \otimes \left(c_a^- |E_a^-\rangle + c_a^+ |E_a^+\rangle \right) + |\xi\rangle$$

- $E_a^+ - E_a \leq \epsilon_{\text{prec}}, \quad E_a - E_a^- \leq \epsilon_{\text{prec}}$
- $|c_a^+|^2 + |c_a^-|^2 \geq 1 - \epsilon_{\text{fail}}$
- $\|\xi\|^2 \leq \epsilon_{\text{fail}}$

After quantum phase estimation, we apply the rotation controlled by the value in the energy register to prepare the state $|\tilde{\Psi}\rangle$

$$\frac{1}{\sqrt{D}} \sum_{a=1}^D |\psi_a\rangle \otimes |\bar{\psi}_a\rangle \otimes c_a^\pm |E_a^\pm\rangle \otimes \left(\sqrt{e^{-\beta E_a^\pm}} |0\rangle + \sqrt{1 - e^{-\beta E_a^\pm}} |1\rangle \right) + |\tilde{\xi}\rangle$$

Amplitude amplification

We apply amplitude amplification to obtain

$$|\tilde{\beta}\rangle := \frac{\Pi_0|\tilde{\Psi}\rangle}{\|\Pi_0|\tilde{\Psi}\rangle\|}$$

The required number of iterations is

$$1/\|\Pi_0|\tilde{\Psi}\rangle\|$$

The desired approximate thermal state is

$$\tilde{\rho}_\beta = \text{Tr}_{\bar{A}}(|\tilde{\beta}\rangle\langle\tilde{\beta}|)$$

where we trace out over the ancilla, the energy register and the register holding $|\bar{\psi}_a\rangle$

The approximate thermal state $\tilde{\rho}_\beta$ is close to the perfect ρ_β provided that simulation and phase estimation are accurate enough

Given H , β , and ϵ , our quantum algorithm makes it possible to prepare $\tilde{\rho}_\beta$

$$\|\rho_\beta - \tilde{\rho}_\beta\|_{\text{tr}} \leq \epsilon$$

The expected run time is

$$\sqrt{\frac{D}{Z_\beta}} \cdot \frac{\beta}{\epsilon} \cdot \left(\log \frac{1}{\epsilon} + \beta \right)$$